Engineering Notes — ApTap Release 2

1. EDA, ingestion and data quality

Verified locally, 15 October 2026: an isolated fresh July–September load reconciled **135,155 raw = 133,692 active + 1,463 rejected**; 1,423 canonical duplicates and 40 missing customer IDs, with no canonical corrections. The pipeline took **8.842 s / 64.4 MiB peak RSS** on one Mac. Execution/test evidence and reproducible commands are in release readiness; these measurements are not a production benchmark.

July/August signed-amount CSV with ISO dates migrates to September JSONL: renamed IDs, day-first `posted`, positive GBP `value` and `direction`. **22,476** September values contain `£`; applying direction after parsing prevents sign errors. Schema-specific adapters normalise to Silver; additive JSON fields are accepted, unsupported filenames/contracts fail, and invalid required fields, dates, amounts or currencies are audited. August includes **1,052 July bookings**: delivery period must remain separate from booking period. **184 raw descriptions become 92 normalised descriptions**; raw bytes/narratives remain in source/Bronze rather than being lost to matching normalisation.

September has **312 repeat-ID occurrences with different raw JSON payloads**; these are formatting variations, not evidence of canonical corrections. Bank-scoped IDs plus canonical payload hashes distinguish duplicates from genuine changes; never deduplicate by amount/date/description alone. Identical canonical occurrences enter the rejection ledger. A valid changed delivery supersedes active Silver with original-payload audit and correction lineage; invalid resends cannot replace it. **Latest valid observed delivery wins**, using filename/row order initially and committed arrival order later: without source versions this is an assumption, not proof of authority. Immutable content-addressed Bronze, file manifests and reconciliation account for every occurrence; identical file digests are skipped. October discovery uses the filename period and automatically advances the cutoff; the actual employer October file remains untested.

2. Classification choices and what remains untrusted

There are only **57 labelled descriptions**, with one gym and two home-insurance examples; repeated transactions add no independent ground truth. `OCTOPUS ENERGY` and `TESCO MOBILE` have incorrect source labels, corrected separately without altering raw labels. Complex newly trained supervised models would risk overfitting this sparse, imbalanced sample. Character TF-IDF provides inexpensive offline retrieval; versioned curated aliases provide auditable demo decisions. **Aliases are synthetic demo assumptions, not bank-approved product truth**: multi-service insurer/telecom names need independent review.

Runtime acceptance comes from demo aliases; otherwise a lexical candidate receives one of exactly ten scored categories plus `needs_review=true`, and is excluded from trusted business features. `REVIEW_REQUIRED` is a workflow status; `not_a_bill` is a category. Optional pretrained local MiniLM supplies **non-voting diagnostics**, not runtime approval; similarity or agreement is not calibrated correctness. No inference-time network calls are needed.

The isolated exploratory leave-one-description-out lexical evaluation gives **accuracy 0.789, macro-F1 0.635, macro-AP 0.698**. Macro-F1 exposes minority-class errors; per-class precision/recall and PR/AP reveal ranking trade-offs. This is neither an independent holdout nor production precision. Earlier MiniLM scores remain historical, not rerun here. **88.76% accepted coverage is not accuracy**; 21/92 descriptions (15,033 transactions) require review. Two accepted Admiral aliases expose another 851 transactions to governance risk; revocation remains a proposal, not an implemented override.

6,681 bills and 209 switches are candidates, not confirmed contracts/cancellations. Bills require current-month observations; 120 have insufficient recurrence history. Validators check completeness, supporting Silver IDs, cadence/dates, individual-payment medians and provider histories. Same-day equal payments with distinct IDs are flagged, not silently deleted. Switch evidence is 87 provisional and 122 repeated-new-provider patterns, with zero externally confirmed switches or verified savings. The short observational window, selection and plan/price confounding prevent causal or annual savings claims. The report retains chart-based uncertainty and category sample limitations.

3. Proposed monthly production in two settings

ApTap-managed AWS, only with authorised transfer: encrypted resumable export, checksums and atomic READY manifest → private S3 immutable Bronze → EventBridge/Step Functions → Glue/EMR validation and quarantine → Iceberg Silver `MERGE`/correction audit → incremental customer-period Gold → pinned offline classifier and review registry → reports. Use IAM least privilege, KMS, private networking, bank isolation, audit, retention/deletion and data-quality/cost alerts. Incomplete multipart deliveries must never become complete batches.

Fully bank-contained: approved chunked/resumable handoff with per-chunk/full hashes and local atomic manifest → bank-owned object storage → approved Spark/SQL scheduler and Bronze/Silver/Gold → signed offline classifier or approved native equivalent → bank-local review/dashboard. Resume verified chunks after timeouts. No customer data, model queries or telemetry leave; no arbitrary package installs or public weight downloads. Dependencies/SBOM, models, aliases, testing, release and rollback pass bank change control. Any exception for de-identified aggregate egress requires explicit bank approval.

Scale/storage: September's 44,950 records/1,500 customers extrapolate to **449.5m rows/month, ~64.6 GB raw JSONL** at 15m customers, before replicas, columnar copies and audit. Applying the fresh three-month throughput gives **8.842 × 449.5m / 135,155 ≈ 8.17 h** for one such month; this ignores skew, contention, I/O and parallelism. Linear RSS extrapolation gives roughly **209 GiB**, equally unsuitable as a capacity requirement: measure bounded distributed partitions instead. SQLite's single writer, global lookups, full-history recomputation and failed large transfers break first. Partition by bank/booking month, retain delivery lineage, cluster by customer/date, index bank+transaction ID, compact files, and recompute affected windows only. AWS, bank deployment and Spark/Iceberg CDC parity are **unverified proposals**.

Privacy/cost: customer/transaction references, narratives, dates, amounts and spending patterns can identify people. Minimise/tokenise appropriately, encrypt, restrict access, redact operational logs and apply contractual retention/erasure. Only authorised data may enter ApTap AWS; bank-contained raw data stays local. Measure compute, storage, requests, transfer, orchestration, security, inference and review cost per actual million transactions. The archived Glue **$0.063/million compute-only scenario uses assumed price/throughput**; no measured cloud or all-in cost exists.

4. Proposed MLOps, deterioration and first week

Establish a versioned, independently adjudicated merchant-level reference set, stratify rare/new merchants, keep a hidden holdout, and report per-class support and uncertainty. Compare majority-class and alias-rule baselines with TF-IDF and an optional MiniLM challenger. Pool months if label support is inadequate; **0.635 exploratory F1 is not a production acceptance baseline**. Monitor delivery/schema drift, reconciliation, corrections, category mix, unseen descriptors, review backlog, sampled accepted precision, false-switch cases, latency and cost.

Investigate an adequately supported **10% relative macro-F1 decline** from a validated baseline (0.70 to 0.63), confirmed over two windows or an uncertainty-adjusted material fall. Proposed PSI warnings **≥0.10**, investigation **≥0.20**, on stable amount/text-length/known-merchant bins and category mix need bank/seasonality calibration and zero-bin safeguards. **PSI measures distribution shift, not accuracy**. Neither gate is implemented. Diagnose source/policy issues, collect labels and correct the registry first; train challengers only with sufficient new ground truth, check untouched holdout and high-risk/minority errors, then approve, sign, canary and retain rollback. No automatic promotion or scheduled retraining is justified.

First week: agree security, contracts and correction order; establish merchant-product review; exercise October-style replay/backfill and interrupted uploads; benchmark representative distributed slices and skew; collect independent labels; baseline monitoring, runtime, memory and all-in cost before agreeing an SLA. Hidden October performance, authoritative corrections, true switches, production precision and scale remain unknown.

AI use: ChatGPT/coding assistance supported interpretation, Python, tests, debugging, architecture and writing. Every complete original conversation/export, applicable share link and coding-agent history (including these audits) is still required. Reconstructions are supplementary. Both previous Notes versions are preserved in the archive; see AI status.